

Rigid Motions on the Coordinate Plane

Answer Key

High School Geometry

Quick Answers

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|---|--|
| 1. x-coordinate of $A' = 2$, y-coordinate of $A' = -2$ | 6. x-coordinate of $E' = 10$, y-coordinate of $E' = 4$ |
| 2. x-coordinate of $B' = -2$, y-coordinate of $B' = -5$ | 7. (a) x-coordinate of $A'' = -2$, (a) y-coordinate of $A'' = 6$, (b) $AB = 4$, (c) why the image length is unchanged = — |
| 3. x-coordinate of $C' = 6$, y-coordinate of $C' = -3$ | 8. (a) $EF = 10$, (b) x-coordinate of $E' = 5$, (b) y-coordinate of $E' = 2$, (c) $E'F' = 10$, (d) why every rigid motion preserves the length = — |
| 4. (a) x-coordinate of $P' = 1$, (a) y-coordinate of $P' = -4$, (b) $PQ = 5$, (c) $P'Q' = 5$ | |
| 5. (a) x-coordinate of $D' = -5$, (b) $DD' = 14$ | |
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What is verified

21 of 23 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-CO.A – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 23 tagged checks.

- 1.** $A(-4, 1)$ under $(x, y) \rightarrow (x + 6, y - 3)$. Find A' .

Solution: A translation acts on each coordinate independently, so substitute into each half of the rule.

$$x' = -4 + 6 = 2, \quad y' = 1 - 3 = -2$$

So

$A'(2, -2)$

On the grid, A' sits six squares right and three squares down from A — the same slide every point of the figure receives.

- 2.** $B(5, 2)$ reflected in the line $y = -x$. Find B' .

Solution: The rule for reflecting in $y = -x$ is $(x, y) \rightarrow (-y, -x)$: swap the coordinates and negate both.

$$x' = -2, \quad y' = -5$$

So

$$B'(-2, -5)$$

3. $C(-6, 3)$ given a half turn about the origin. Find C' .

Solution: A half turn about the origin is $(x, y) \rightarrow (-x, -y)$ — every point goes to the opposite side of the origin, the same distance away.

$$x' = -(-6) = 6, \quad y' = -3$$

So

$$C'(6, -3)$$

4. $P(-3, 2)$, $Q(1, 5)$ translated by $(x, y) \rightarrow (x + 4, y - 6)$; $Q'(5, -1)$.

Solution (a): $x' = -3 + 4 = 1$ and $y' = 2 - 6 = -4$, so

$$P'(1, -4)$$

Solution (b): $PQ = \sqrt{(1 - (-3))^2 + (5 - 2)^2} = \sqrt{16 + 9} = \sqrt{25}$, so

$$PQ = 5$$

Solution (c): $P'Q' = \sqrt{(5 - 1)^2 + (-1 - (-4))^2} = \sqrt{16 + 9} = \sqrt{25}$, so

$$P'Q' = 5$$

The horizontal and vertical gaps between the two points are unchanged by the slide, which is exactly why the two lengths agree.

5. $D(9, 4)$ reflected in the line $x = 2$. (a) The x -coordinate of D' . (b) DD' .

Solution (a): Reflecting in the vertical line $x = c$ sends x to $2c - x$; the mirror line sits halfway between a point and its image.

$$x' = 2(2) - 9 = 4 - 9$$

So

$$-5$$

Solution (b): $D(9, 4)$ and $D'(-5, 4)$ share a y -coordinate, so the distance is the horizontal gap: $|9 - (-5)| = 14$.

$$DD' = 14$$

That is twice the 7-unit gap from D to the mirror line, as a reflection always gives.

6. $E(3, -7)$ turned a quarter turn counterclockwise about $(1, 2)$. Find E' .

Solution: With centre (h, k) , the rule is $(x, y) \rightarrow (h - (y - k), k + (x - h))$. Here $h = 1$ and $k = 2$.

$$x' = 1 - (-7 - 2) = 1 + 9 = 10$$

$$y' = 2 + (3 - 1) = 2 + 2 = 4$$

So

$$E'(10, 4)$$

Check: E is 2 right and 9 down from the centre; a quarter turn counterclockwise sends "2 right, 9 down" to "9 right, 2 up", which lands exactly on $(10, 4)$ ✓

7. $A(2, 1)$, $B(6, 1)$, $C(6, 4)$ reflected in the y -axis, then translated by $(x, y) \rightarrow (x, y + 5)$.

Solution (a): Apply the two rules in order. The reflection sends $A(2, 1)$ to $(-2, 1)$; the translation then adds 5 to the y -coordinate.

$$A'' = (-2, 1 + 5)$$

So

$$A''(-2, 6)$$

Solution (b): A and B share a y -coordinate, so $AB = |6 - 2|$.

$$AB = 4$$

Solution (c) — instructor-judged. A model response: a reflection is a rigid motion and a translation is a rigid motion, and a rigid motion by definition preserves every distance. Doing one and then the other preserves distance twice over, so $A''B''$ must still measure 4 — no computation of B'' is needed.

Full credit: names both steps as rigid motions and states the distance-preserving property. *Half credit:* asserts the length is preserved without naming both steps. *No credit:* recomputes the distance instead of reasoning.

8. $E(-5, -2)$, $F(3, 4)$, both given a half turn about the origin; $F'(-3, -4)$.

Solution (a): $EF = \sqrt{(3 - (-5))^2 + (4 - (-2))^2} = \sqrt{64 + 36} = \sqrt{100}$, so

$$EF = 10$$

Solution (b): A half turn about the origin negates both coordinates: $x' = 5$ and $y' = 2$,

so

$$E'(5, 2)$$

Solution (c): $E'F' = \sqrt{(-3 - 5)^2 + (-4 - 2)^2} = \sqrt{64 + 36} = \sqrt{100}$, so

$$E'F' = 10$$

Solution (d) — instructor-judged. A model response: a rigid motion is a transformation of the plane that preserves distance. Translations slide the plane, reflections flip it, and rotations turn it — none of them stretches, shrinks or bends it, so the gap between any two points is carried over unchanged. The two lengths in parts (a) and (c) therefore had to come out equal; the agreement is a consequence of the definition, not a coincidence.

Full credit: states the distance-preserving definition and applies it to the three motions. *Half credit:* names the motions without the property. *No credit:* offers the two equal numbers as the reason.